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Pilot-Scale Hydrodynamic Cavitation Pretreatment for Improving Anaerobic Digestion of Cattle Manure: Methane Kinetics, Energy Performance and Digestate Quality

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Abstract: The recalcitrant particulate structure of cattle manure limits hydrolysis during anaerobic digestion, thereby reducing methane productivity and renewable energy recovery in agricultural biogas systems. This study investigated pilot-scale hydrodynamic cavitation (HC) pretreatment of cattle manure followed by laboratory-scale batch anaerobic digestion, using an integrated Venturi-based cavitation reactor and a programmable logic controller–supervisory control and data acquisition (PLC–SCADA) process monitoring system. The approach combined mechanical pretreatment, HC and automated process monitoring to improve substrate conversion efficiency, digestion stability and overall process performance. Methane production kinetics were evaluated using the modified Gompertz model, while digestate nutrient composition and energy performance were also assessed. HC increased cumulative methane production from 234.9 ± 5.9 to 324.7 ± 7.0 mL g^{-1} COD, corresponding to an increase of 38.2%, while cumulative biogas production increased from 397.6 ± 7.1 to 497.7 ± 6.5 mL g^{-1} COD (25.2%). Peak daily methane production was higher in the cavitation-treated reactors, consistent with enhanced methane-production kinetics following HC pretreatment. Modified Gompertz modeling confirmed higher methane production potential (239.5 → 330.8 mL g^{-1} COD) and maximum methane production rate (35.3 → 48.9 mL g^{-1} COD d^{-1}), whereas the lag phase remained short in both treatments. HC also increased total nitrogen (N), phosphorus expressed as P_2O_5, and potassium expressed as K_2O contents in the digestate on a dry-weight basis. Relatively stable pH conditions were maintained throughout digestion, with lower pH variability observed in the HC-treated reactors. Energy analysis showed a positive incremental net energy balance, with an additional methane energy recovery of 42.97 kWh m^{-3}, a specific additional electrical energy input of 19.0 kWh m^{-3}, and an incremental net energy ratio (NER) of 2.26. Overall, the results show that HC can enhance methane production and methane-production kinetics and may support nutrient recycling while maintaining positive incremental energy performance. The integrated cavitation and PLC–SCADA system offers a promising approach for sustainable livestock waste management in organic farming systems.

Keywords: Anaerobic digestion; Cattle manure; Hydrodynamic cavitation; Methane production; Biogas; Digestate; Kinetic modeling; Circular bioeconomy.

1. Introduction

Livestock manure management remains a major environmental and technological challenge due to increasing volumes of organic waste generated by intensive agricultural production (Manyi-Loh et al., 2013; Song et al.,

2023). Conventional manure handling practices are associated with greenhouse gas emissions, nutrient losses, odor generation, and contamination of soil and water resources. In this context, anaerobic digestion has attracted considerable attention as a sustainable technology capable of simultaneously recovering renewable energy in the form of biogas and producing nutrient-rich digestate suitable for agricultural applications (Mao et al., 2015; Möller & Müller, 2012; Song et al., 2023). Consequently, anaerobic digestion is increasingly considered an important component of circular bioeconomy strategies aimed at improving waste valorization and reducing the environmental footprint of livestock farming.

Hydrolysis is generally considered the rate-limiting stage of anaerobic digestion, particularly for lignocellulosic and particulate substrates such as cattle manure (Hendriks & Zeeman, 2009; Zheng et al., 2014). The high content of cellulose, hemicellulose, lignin, and fibrous organic matter limits substrate biodegradability and slows microbial conversion of complex organic compounds into soluble intermediates. As a consequence, untreated cattle manure typically exhibits relatively low methane productivity and requires extended hydraulic retention times to achieve stable biogas generation.

To overcome these limitations, various pretreatment approaches have been investigated, including thermal, chemical, mechanical, ultrasonic, and biological methods (Ariunbaatar et al., 2014; Carlsson et al., 2012). These pretreatments aim to improve substrate solubilization and increase the accessibility of organic matter to hydrolytic and methanogenic microorganisms. However, many conventional pretreatment technologies remain energy-intensive, operationally complex, or difficult to implement under industrial agricultural conditions.

Among emerging pretreatment technologies, hydrodynamic cavitation (HC) has attracted increasing attention as a promising method for intensification of anaerobic digestion processes (Dębowski et al., 2024; Langone et al., 2018; Patil et al., 2016). During cavitation, the collapse of vapor bubbles generates localized shear forces, microjets, and pressure fluctuations capable of disrupting biomass structure and fragmenting particulate organic matter (Dular et al., 2016; Ranade, 2022). Such effects may enhance substrate solubilization and improve accessibility of organic compounds to anaerobic microorganisms. Previous studies have demonstrated that HC pretreatment can increase methane yield, accelerate hydrolysis, and improve biogas productivity during digestion of livestock waste and other organic substrates (Dębowski et al., 2024; Langone et al., 2018; Patil et al., 2016).

Despite the promising results reported for HC-assisted anaerobic digestion, several important limitations remain insufficiently addressed. Most published studies have been conducted under laboratory-scale conditions, which do not fully reflect the hydraulic, energetic and operational complexity of industrial biogas systems (Coarita Fernandez et al., 2020; Patil et al., 2016). Although several investigations have progressed towards pilot-scale or industrial-scale validation (Dębowski et al., 2024; Langone et al., 2018), comprehensive evaluations integrating methane production, process stability, kinetic behavior, digestate quality and energy performance within a single experimental framework remain relatively uncommon.

Another important limitation is the relatively low level of automation and adaptive process control implemented in most cavitation-assisted digestion systems. Real-time regulation of pressure, flow conditions, and pretreatment intensity remains poorly investigated, particularly under variable substrate conditions typical for agricultural waste streams. Furthermore, comprehensive energy assessments and integrated evaluations of methane productivity, digestion stability, and nutrient recovery are still limited (Batstone et al., 2015; Ge et al., 2021; Ranade, 2022; Uddin & Wright, 2023).

Therefore, there remains a need for studies that integrate pilot-scale HC pretreatment, automated process monitoring, kinetic modeling, digestate quality evaluation and comprehensive energy-performance assessment within a unified anaerobic digestion framework (Dębowski et al., 2024; Uddin & Wright, 2023). Such integrated approaches are particularly relevant for organic and sustainable farming systems, where livestock manure valorization simultaneously supports renewable energy production and nutrient recycling through the application of digestate as an organic fertilizer (Möller & Müller, 2012; Nkoa, 2014).

Recent reviews have highlighted the increasing importance of process intensification, pretreatment optimization and integrated monitoring in anaerobic digestion systems treating agricultural residues and livestock manure (Khoshnevisan et al., 2021; Uddin & Wright, 2023).

In this context, the present study investigates pilot-scale HC pretreatment of cattle manure followed by laboratory-scale batch anaerobic digestion using an integrated cavitation and process automation system. The study evaluates methane production kinetics using the modified Gompertz model and investigates the influence of cavitation pretreatment on digestate nutrient composition and overall energy performance.

Within organic farming systems, efficient livestock manure management is closely linked to both renewable energy recovery and nutrient recycling. Improving anaerobic digestion efficiency not only increases biogas production but also enhances the utilization of digestate as an organic fertilizer, thereby supporting circular nutrient management and reducing environmental losses associated with manure storage. Consequently, technologies that simultaneously improve methane recovery and manure valorization directly contribute to the sustainability objectives of modern organic farming.

The novelty of the present study lies not in the individual application of HC or anaerobic digestion, both of which have been previously investigated, but in their integrated evaluation within a pilot-scale automated

processing system. Unlike earlier studies that primarily focused on methane enhancement under laboratory conditions, the present work combines Venturi-type HC, continuous programmable logic controller–supervisory control and data acquisition (PLC–SCADA) process monitoring, kinetic analysis using the modified Gompertz model, comprehensive digestate characterization, and an experimentally based energy performance assessment within a single experimental framework. This integrated approach enables simultaneous evaluation of biological performance, process stability, nutrient recycling potential, and energy efficiency through pilot-scale pretreatment and laboratory-scale anaerobic digestion experiments, thereby providing a more comprehensive assessment of the practical applicability of HC for sustainable livestock manure management.

2. Materials and Methods

2.1 Substrate Characteristics

Fresh cattle manure used in the present study was obtained from the Akzhar Farm located in the Zhambyl Region, Kazakhstan. The substrate consisted of manure freshly collected from livestock housing facilities and transported immediately to the pilot-scale HC pretreatment unit. All physicochemical analyses were performed prior to HC pretreatment.

Before cavitation treatment, the manure was mechanically homogenized to reduce particle-size heterogeneity and ensure uniform substrate loading during the anaerobic digestion experiments. The substrate contained approximately 85–88% moisture, corresponding to a total solids (TS) content of 12–15% on a fresh-weight basis (FW), while the initial pH ranged between 6.7 and 6.9, indicating conditions suitable for subsequent anaerobic digestion. The initial physicochemical characteristics of the cattle manure are summarized in Table 1.

Table 1. Initial physicochemical characteristics of cattle manure used in the anaerobic digestion experiments

Parameter	Unit	Mean ± SD
Moisture content	% FW	85.92 ± 0.24
Total solids (TS)	% FW	14.08 ± 0.24
Volatile solids (VS)	% FW	11.68 ± 0.26
VS/TS ratio	%	82.95 ± 0.29
pH	Dimensionless	6.80 ± 0.10
Total chemical oxygen demand (COD)	g L^{-1}	48.17 ± 0.63
Total nitrogen (N)	% DW	1.62 ± 0.03
Phosphorus (as P_2O_5)	% DW	0.84 ± 0.03
Potassium (as K_2O)	% DW	1.11 ± 0.03

Note: SD = standard deviation; FW = fresh weight; DW = dry weight.

Chemical oxygen demand (COD) was determined using the closed-reflux dichromate method in accordance with ISO 15705:2002 (small-scale sealed-tube method) (ISO, 2002). Owing to the high organic content of cattle manure, samples were diluted with deionized water prior to digestion to bring the COD concentration within the analytical range of the method. The dilution factor was recorded for each sample and applied when calculating the COD concentration of the original undiluted sample.

TS were determined by drying the samples at 105 °C to constant weight in accordance with GOST 26713-85 (USSR State Committee for Standards, 1985a). Volatile solids (VS) were determined by subsequent ignition of the dried residue at 550 ± 50 °C in a muffle furnace according to Standard Methods 2540 E (American Public Health Association et al., 2017).

All physicochemical analyses were performed in triplicate, and the results are reported as mean ± standard deviation (SD).

The fresh cattle manure exhibited a total COD of 48.17 ± 0.63 g L^{-1}, a TS content of 14.08 ± 0.24% FW and a VS content of 11.68 ± 0.26 % FW. The corresponding VS/TS ratio of 82.95 ± 0.29% indicates that the substrate contained a high proportion of biodegradable organic matter, suggesting its suitability for anaerobic digestion experiments and providing an appropriate substrate for evaluating HC pretreatment performance.

2.2 Hydrodynamic Cavitation Pretreatment System

HC pretreatment was carried out using a pilot-scale raw material processing unit (RMPU) integrating mechanical pretreatment, HC and PLC–SCADA-based process monitoring. The pretreatment line consisted of a fecal pump, screw shredder, knife shredder, macerator, gerotor pump, Venturi-type cavitation reactor, heating unit and programmable control module. A schematic representation of the pilot-scale RMPU integrating mechanical pretreatment, HC and PLC–SCADA process control is presented in Figure 1.

HC was generated in a Venturi-type cavitation reactor, consisting of an inlet section, converging section, throat, cavitation zone and diverging outlet. During operation, the acceleration of the substrate within the throat produced

a local pressure reduction below the vapor pressure of the liquid, resulting in cavitation bubble formation. Subsequent pressure recovery downstream of the throat induced rapid bubble collapse, generating localized pressure pulses, microjets and shear forces responsible for substrate disintegration. The configuration of the Venturi reactor and the location of the cavitation zone are schematically illustrated in Figure 2.

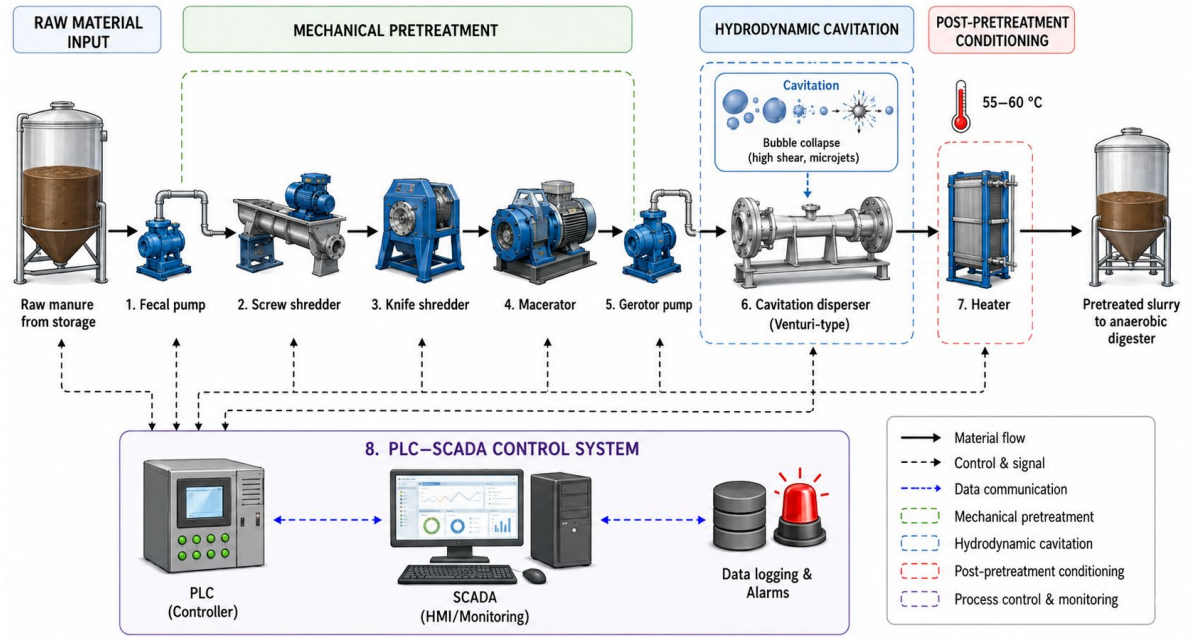


Figure 1. Schematic representation of the pilot-scale RMPU integrating mechanical pretreatment, HC and PLC–SCADA process control

Note: RMPU = raw material processing unit; HC = hydrodynamic cavitation; PLC–SCADA = programmable logic controller–supervisory control and data acquisition.

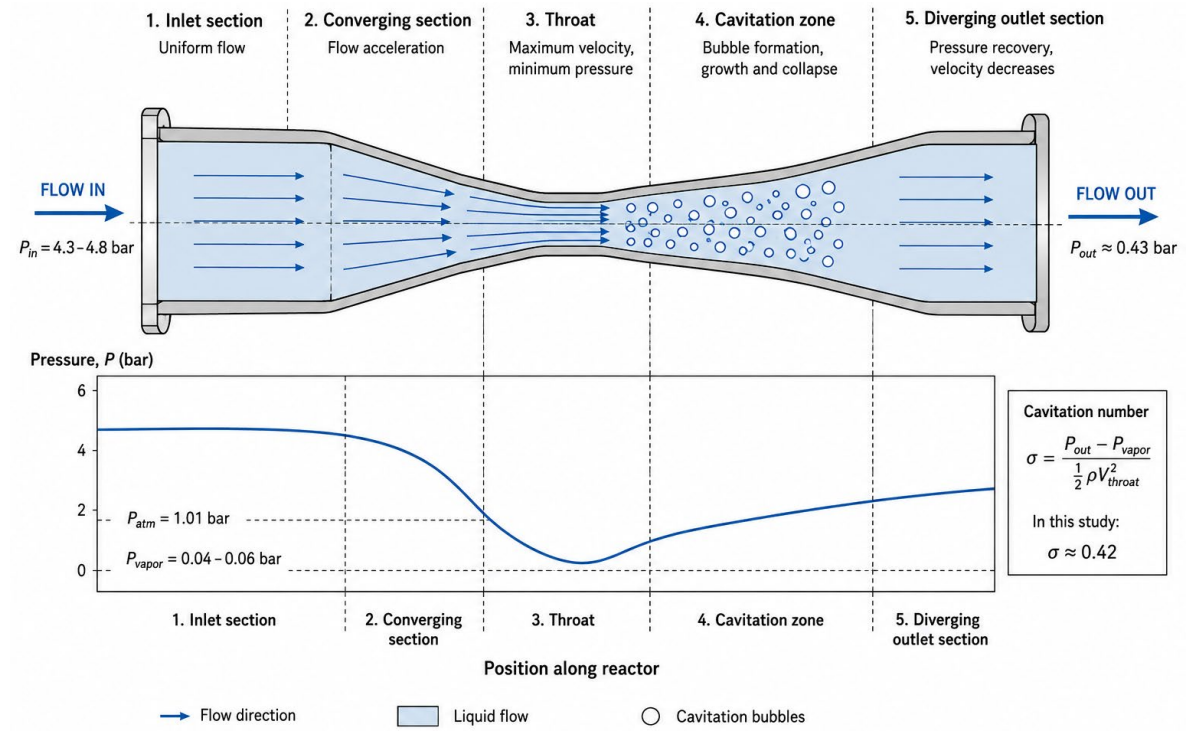


Figure 2. Schematic illustration of the Venturi-type HC reactor and cavitation zone formation

Note: HC = hydrodynamic cavitation; P_{in} = inlet pressure; P_{out} = downstream (outlet) pressure; P_{vapor} = vapor pressure of the liquid; ρ = slurry density; v_{throat} = average velocity at the Venturi throat; σ = cavitation number.

The mechanisms governing cavitation bubble formation and collapse in Venturi reactors have been extensively

described in previous studies (Ge et al., 2021). During pretreatment, cattle manure slurry was circulated through the cavitation reactor at a flow rate of approximately 150 L h^{-1}. The inlet and outlet pressures were measured as absolute pressures and ranged from 4.3 to 4.8 bar and 0.40 to 0.46 bar, respectively, corresponding to a pressure drop of 3.9–4.1 bar. Pretreatment temperature was maintained at approximately 45 °C. The cavitation number (σ) was calculated based on the reactor operating conditions and the calculation approach illustrated in Figure 2. Under the investigated conditions, the calculated cavitation number was σ ≈ 0.42. A photographic view of the pilot-scale RMPU used in the pretreatment experiments is shown in Figure 3.

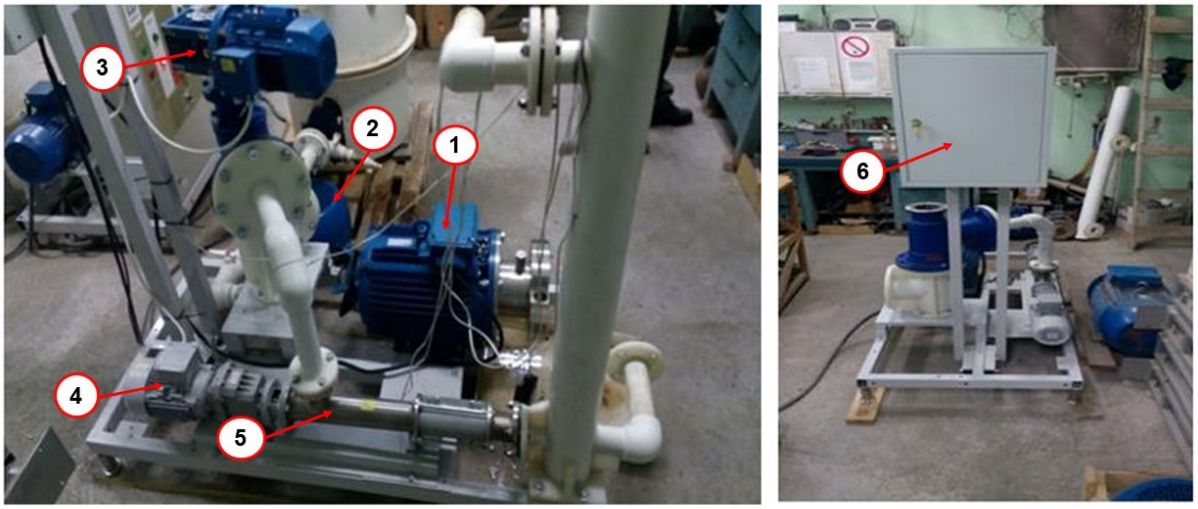


Figure 3. Photographic views of the pilot-scale RMPU: (a) front-side view showing the main pretreatment components; (b) rear-side view showing the PLC–SCADA control cabinet and hydraulic connections
Note: RMPU = raw material processing unit; PLC–SCADA = programmable logic controller–supervisory control and data acquisition. The numbered callouts indicate the principal components of the pretreatment system: 1 — screw shredder; 2 — knife shredder; 3 — macerator; 4 — gerotor pump; 5 — Venturi-type cavitation reactor (cavitation disperser); 6 — PLC–SCADA control panel.

Table 2. Operating parameters of the HC pretreatment

Parameter	Unit	Value
Reactor type	—	Venturi-type cavitation reactor
Flow rate	L h^{-1}	150
Operation time per run	h	0.67
Processed substrate volume	L run^{-1}	100.5
Inlet pressure	bar	4.3–4.8
Outlet pressure	bar	0.40–0.46
Pressure drop	bar	3.9–4.1
Pretreatment temperature	°C	~45
Cavitation number (σ)	Dimensionless	~0.42
Control electrical energy	kWh run^{-1}	3.05
HC electrical energy	kWh run^{-1}	4.96
Additional electrical energy	kWh run^{-1}	1.91
Specific additional electrical energy	kWh m^{-3} substrate	19.0
Inlet diameter	mm	25–32
Throat diameter	mm	8–12
Outlet diameter	mm	25–32
Throat length	mm	10–25
Converging/diverging angles	°	20–25
Reactor material	—	Welded carbon steel (St3), wall thickness 3–5 mm

Note: HC = hydrodynamic cavitation; "—" indicates not applicable. Pressures used for the cavitation number calculation are absolute pressures. The processed substrate volume was estimated from the measured flow rate and pretreatment duration. Electrical energy consumption was determined from the programmable logic controller–supervisory control and data acquisition (PLC–SCADA) energy inventory. The additional electrical energy represents the difference between the HC and control configurations. Geometric characteristics correspond to the engineering specifications of the Venturi reactor used in the pilot-scale pretreatment system.

Each pretreatment cycle lasted 0.67 h, corresponding to a processed substrate volume of approximately 100.5 L (0.1005 m^3). According to the PLC–SCADA energy inventory, the control configuration consumed 3.05 kWh of electrical energy per processing cycle, whereas the HC configuration required 4.96 kWh because of the

additional operation of the gerotor pump and Venturi cavitation reactor. Consequently, the additional electrical energy required for cavitation pretreatment was 1.91 kWh per cycle, corresponding to a specific additional electrical energy input of 19.0 kWh m^{-3} substrate. The principal operating parameters of the HC pretreatment system are summarized in Table 2.

2.3 Anaerobic Digestion Experiments

Batch anaerobic digestion experiments were conducted to compare untreated cattle manure and HC-pretreated cattle manure. Two experimental groups were evaluated: untreated control substrate and HC-pretreated substrate (HC group). Each treatment was performed in triplicate ($n = 3$).

The untreated control and HC-treated groups were processed under identical digestion conditions, differing only in the application of HC pretreatment.

The digestion experiments were performed over a 14-day period under controlled mesophilic conditions. Temperature monitoring showed that the mesophilic digestion zone was maintained at approximately 37 ± 1 °C during the experimental period. The untreated control substrate was not subjected to HC pretreatment, whereas the HC-treated substrate was prepared using the pilot-scale RMPU under the operating conditions described in Section 2.2.

The experimental design was developed with reference to commonly accepted biomethane potential (BMP) testing guidelines proposed by Angelidaki et al. (2009) and Holliger et al. (2016). The anaerobic inoculum used in the batch experiments was obtained from a mesophilic municipal wastewater treatment plant digester (Astana, Kazakhstan), operating at 37 ± 1 °C. The inoculum was transported to the laboratory in sealed containers under anaerobic conditions and was acclimatized to the experimental temperature for 48 h prior to use. The inoculum was collected freshly on the day of experiment setup and was screened through a 2 mm sieve to remove coarse particles and debris. Prior to use, the inoculum was pre-incubated at 37 °C for 48 h under anaerobic conditions to deplete residual biodegradable organic matter and to minimize background methane production during the assays.

The TS and VS content of the inoculum were determined gravimetrically according to GOST 26713-85 (USSR State Committee for Standards, 1985a) by drying the samples at 105 °C to constant weight, followed by ignition at 550 °C to determine volatile solids. The inoculum had a TS content of 4.2 ± 0.3% and a VS content of 2.8 ± 0.2% (fresh weight), corresponding to a VS/TS ratio of approximately 67%. The specific methanogenic activity of the inoculum was not determined. However, blank and positive control assays exhibited stable and reproducible methane production throughout the experimental period, confirming that the inoculum remained biologically active under the applied experimental conditions.

Batch anaerobic digestion experiments were conducted in 500 mL glass serum bottles with a working volume of 350 mL. The remaining headspace volume was approximately 150 mL. Each bottle was filled with a mixture of substrate, inoculum, and distilled water to achieve the desired working volume. The inoculum-to-substrate ratio (ISR) was set at 2:1 on a VS basis (g VS inoculum g^{-1} VS substrate), which is within the recommended range (ISR ≥ 2) to ensure sufficient buffering capacity and to avoid substrate overloading. For the HC-treated group, the substrate was added after cavitation pretreatment, while the control group received untreated substrate. The VS concentrations in the final digestion mixture were approximately 5 g VS L^{-1} for the substrate and 10 g VS L^{-1} for the inoculum, calculated based on the final working volume, resulting in a total VS concentration of approximately 15 g VS L^{-1}.

Blank assays were prepared identically to the substrate assays but without the addition of substrate, containing only inoculum and distilled water. The methane production from the blank assays was subtracted from the gross methane production of the substrate assays to correct for the endogenous methane production of the inoculum. Positive control assays using microcrystalline cellulose (Avicel® PH-101) were also included to verify the methane-producing performance of the inoculum. All assays were performed for each experimental condition in accordance with the validation criteria for BMP tests.

After filling, the bottles were sealed with butyl rubber stoppers and aluminum crimp caps. The headspace was flushed with nitrogen gas (99.999% purity) for 3 min to establish anaerobic conditions. The bottles were then placed in a temperature-controlled incubator at 37 ± 1 °C and were manually shaken twice daily for 1 min to ensure proper mixing and mass transfer.

The pretreatment process was operated using the PLC–SCADA system, whereas the anaerobic digestion experiments were conducted under controlled laboratory conditions. Temperature and pH were measured periodically throughout the digestion period. Methane concentration and cumulative biogas production were measured daily. Methane production kinetics were evaluated using cumulative methane production profiles and daily methane production rates. Process stability was assessed based on pH dynamics.

Cumulative methane production was calculated from the measured biogas volume and methane concentration. All gas volumes were corrected to standard temperature and pressure (STP: 0 °C and 1 atm) according to the ideal gas law. Both cumulative methane production and cumulative biogas production were expressed using the same normalization basis (mL g^{-1} COD) to facilitate direct comparison of treatment performance. Normalization was

performed using the experimentally determined initial COD content of fresh cattle manure (48.2 ± 0.6 g L^{-1}), as presented in Table 1.

2.4 Measurement Procedures

Substrate and digestate characterization included determination of COD, pH, methane concentration, cumulative biogas production, and nutrient composition (nitrogen, phosphorus, and potassium). Methane production and cumulative biogas yield were monitored throughout the anaerobic digestion experiments.

The pH measurements were performed using calibrated digital pH sensors integrated into the PLC–SCADA monitoring system. Temperature was measured using PT100 sensors with an accuracy of ±0.2 °C, while pressure measurements were conducted using calibrated digital pressure sensors with an accuracy of ±0.05 bar.

Total nitrogen (N) was determined by the titrimetric method after distillation according to ISO 5315:1984 (ISO, 1984). Total phosphorus was determined by the photometric method according to GOST 26717-85 (USSR State Committee for Standards, 1985b) and expressed as P_2O_5. Total potassium was determined by the flame-photometric method according to GOST 26718-85 (USSR State Committee for Standards, 1985c) and expressed as K_2O. All analyses were performed in triplicate.

Prior to each experimental series, all sensors and analytical instruments were calibrated according to manufacturer recommendations and standard analytical procedures. Analytical measurements were performed according with relevant ISO and GOST standards to ensure data reliability and reproducibility.

2.5 Kinetic Modeling

Methane production kinetics were evaluated using the modified Gompertz model, which is commonly applied for analysis of cumulative methane production during anaerobic digestion processes (Donoso-Bravo et al., 2010).

$$M(t) = P \exp\left\{-\exp\left[\frac{R_m e}{P}(\lambda - t) + 1\right]\right\} \quad (1)$$

where, $M(t)$ represents cumulative methane production at time t; P represents methane production potential (mL g^{-1} COD); R_m represents the maximum methane production rate (mL g^{-1} COD d^{-1}); and λ represents the lag phase duration (days).

The modified Gompertz model was fitted to the mean cumulative methane production curves obtained from the three biological replicates for each experimental group. Thus, the reported kinetic parameters (P, R_m and λ) together with the corresponding goodness-of-fit statistics, including the coefficient of determination (R^2) and root mean square error (RMSE), describe the agreement between the fitted model and the averaged experimental methane production profiles.

Experimental methane production data were fitted to the modified Gompertz equation using nonlinear regression analysis implemented in Python with the SciPy optimization library. The obtained kinetic parameters were used to evaluate the influence of HC pretreatment on substrate biodegradability and methane production dynamics.

2.6 Energy Assessment

Energy performance analysis was conducted to evaluate the incremental energy balance associated with HC pretreatment by comparing the additional methane energy recovered with the additional electrical energy required for cavitation operation. Electrical energy consumption was determined from the energy demand of the pump, mechanical pretreatment units, cavitation module and auxiliary equipment during each experimental run.

Methane energy recovery was estimated from cumulative methane production data using the lower heating value (LHV) of methane (9.97 kWh m^{-3}, equivalent to 35.9 MJ m^{-3}). Energy calculations were normalized to the processed substrate volume under pilot-scale operating conditions.

The total net energy ratio (NER) was calculated as:

$$NER_{\text{total}} = \frac{E_{\text{CH}_4,\text{total}}}{E_{\text{input,total}}} \quad (2)$$

where, $E_{\text{CH}_4,\text{total}}$ represents the total methane energy recovered, $E_{\text{input,total}}$ is the total electrical energy consumed by the treatment configuration.

To specifically evaluate the energy contribution of HC pretreatment, the incremental NER was calculated as:

$$NER_{\text{incremental}} = \frac{\Delta E_{\text{CH}_4}}{\Delta E_{\text{input}}} \quad (3)$$

where, ΔE_{CH_4} is the additional methane energy obtained due to HC pretreatment compared with the untreated control, and ΔE_{input} is the additional electrical energy required for the HC configuration relative to the control configuration.

The obtained energy assessment was intended to provide a preliminary evaluation of process energy performance under pilot-scale operating conditions.

The energy assessment considered only the additional electrical energy required for HC pretreatment and the corresponding additional methane energy recovered under the investigated operating conditions. Energy consumption associated with substrate heating, digester temperature maintenance, mixing, biogas upgrading, gas compression, and downstream energy utilization was excluded from the present analysis. Consequently, the calculated incremental NER represents the energy performance of the cavitation pretreatment stage rather than the overall energy balance of the complete anaerobic digestion system.

2.7 Statistical Analysis

All experiments were performed in triplicate (n = 3), and results are presented as mean ± SD. Normality of the replicate-level data was assessed using the Shapiro–Wilk test (Shapiro & Wilk, 1965). No significant deviations from normality were detected for the analyzed replicate-level endpoints ($p > 0.05$); however, given the small number of biological replicates, the normality results were interpreted cautiously.

Statistical differences between the two independent experimental groups were evaluated using two-sided Welch's t-tests, which do not assume equal variances between groups. For variables measured repeatedly over time, including pH, measurements were first averaged within each biological replicate (reactor), and the resulting reactor-level means were used for between-group statistical comparisons (n = 3 per group). Differences were considered statistically significant at $p < 0.05$.

Each statistical comparison was performed for a distinct predefined endpoint rather than for repeated testing of the same response variable. Consequently, adjustment for multiple comparisons (e.g., Bonferroni or Holm correction) was not applied.

Statistical analysis and kinetic model fitting were performed using Python programming tools, including the SciPy (Virtanen et al., 2020) and NumPy (Harris et al., 2020) libraries.

3. Results

3.1 Methane Production Kinetics

HC significantly enhanced methane production during anaerobic digestion of cattle manure. On day 14, cumulative methane production reached 324.7 ± 7.0 mL g^{-1} COD in the HC-treated group, compared with 234.9 ± 5.9 mL g^{-1} COD in the untreated control. This corresponds to a 38.2% increase in cumulative methane yield after HC pretreatment. The difference between the experimental groups was statistically significant ($p < 0.001$).

A summary of the main statistical parameters is presented in Table 3.

Table 3. Main statistical parameters of anaerobic digestion performance under HC pretreatment

Parameter	Control	HC-Treated	Change (%)	*p*-Value
Cumulative methane production (mL g^{-1} COD)	234.9 ± 5.9	324.7 ± 7.0	+38.2	<0.001
Average methane concentration (%)	57.8 ± 0.7	63.8 ± 0.7	+10.4	<0.001
Cumulative biogas production (mL g^{-1} COD)	397.6 ± 7.1	497.7 ± 6.5	+25.2	<0.001
pH	6.75 ± 0.02	6.96 ± 0.02	+3.1	<0.001
Total nitrogen (N) (%)	1.79 ± 0.03	2.25 ± 0.06	+25.7	0.0015
Phosphorus (as P_2O_5) (%)	0.96 ± 0.02	1.18 ± 0.04	+22.9	0.0019
Potassium (as K_2O) (%)	1.27 ± 0.03	1.59 ± 0.04	+25.2	0.0003

Note: HC = hydrodynamic cavitation; COD = chemical oxygen demand.

As shown in Figure 4, the HC-treated substrate exhibited consistently higher cumulative methane production throughout the entire digestion period. Enhanced methane accumulation became evident during the early stages of digestion and remained consistently above the untreated control until the end of the 14-day experiment. The relatively small variability among biological replicates suggests good reproducibility of the experimental procedure and indicates stable reactor performance under both operating conditions. The higher cumulative methane yield obtained after HC pretreatment is consistent with enhanced substrate conversion and methane-production performance under the investigated conditions.

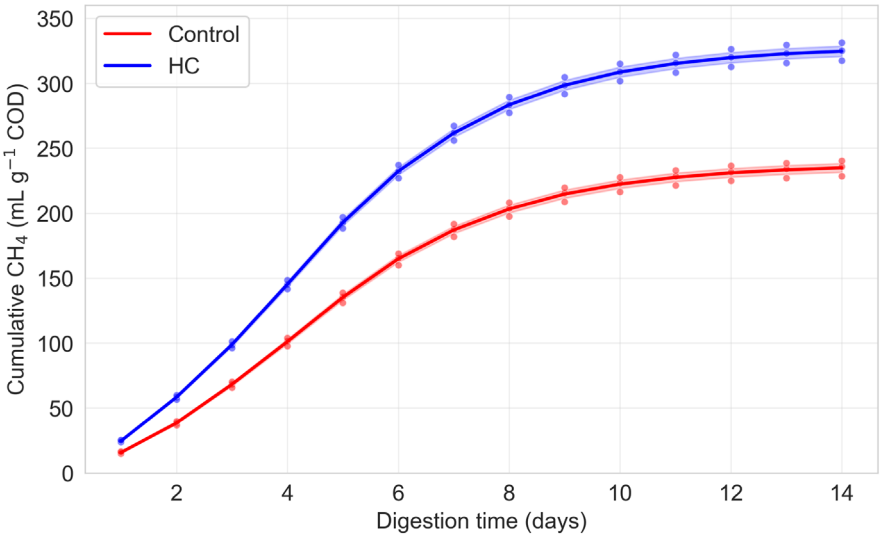


Figure 4. Cumulative methane production during anaerobic digestion of untreated and cavitation-pretreated cattle manure

Note: HC = hydrodynamic cavitation; COD = chemical oxygen demand. Symbols represent individual replicates (n = 3), and solid lines indicate group means, and shaded regions represent the standard error of the mean (SEM).

3.2 Incremental Daily Methane Production Dynamics

Incremental daily methane production, calculated from the differences between consecutive cumulative methane measurements, is presented in Figure 5. Both experimental groups exhibited a progressive increase in daily methane production during the initial digestion stage, reaching maximum production rates on approximately day 5. Thereafter, methane production gradually declined as the readily biodegradable organic fraction became depleted.

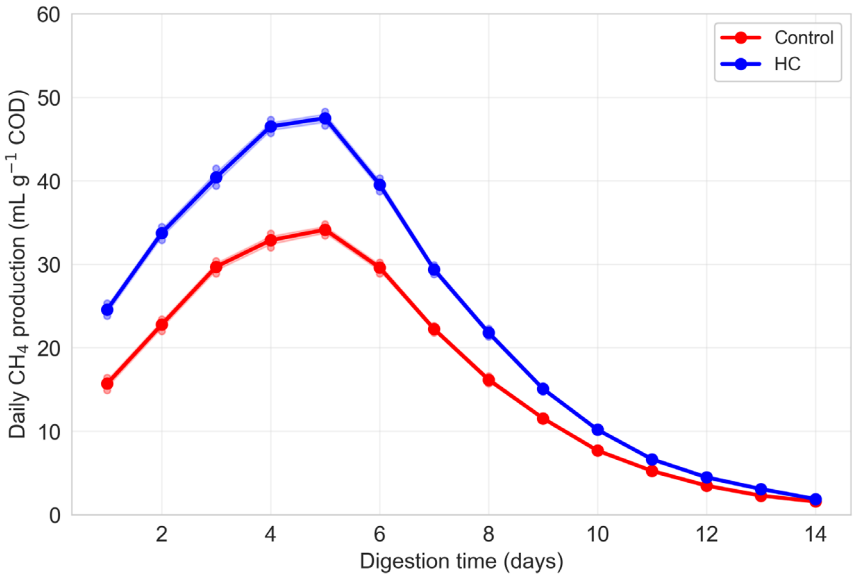


Figure 5. Incremental daily methane production during batch anaerobic digestion of untreated and HC-pretreated cattle manure

Note: HC = hydrodynamic cavitation; COD = chemical oxygen demand. Daily methane production was calculated from the differences between consecutive cumulative methane measurements. Symbols represent individual biological replicates (n = 3), solid lines indicate group means, and shaded regions represent the standard error of the mean (SEM).

Throughout the active digestion period (approximately days 2–9), the HC-treated substrate consistently produced higher daily methane yields than the untreated control. The largest difference between the experimental groups was observed between days 4 and 6, when HC increased the daily methane production rate by approximately 35–40% relative to the untreated substrate. This pattern is consistent with accelerated conversion

of the biodegradable substrate fraction following HC pretreatment; however, hydrolysis was not directly quantified.

Following the peak production stage, both groups exhibited a gradual decrease in daily methane production. However, the HC-treated reactors maintained higher methane production rates than the control throughout the remaining digestion period, indicating sustained enhancement of substrate biodegradability rather than merely an earlier onset of methanogenesis.

These observations are consistent with the cumulative methane production profiles (Figure 4), further supporting the enhancement of methane production following HC pretreatment.

3.3 Process Stability

HC pretreatment maintained stable pH conditions throughout the anaerobic digestion process. The mean reactor-level pH was 6.96 ± 0.02 in the HC-treated reactors and 6.75 ± 0.02 in the untreated control, with the difference between the experimental groups being statistically significant (two-sided Welch's *t*-test, $p < 0.001$; $n = 3$ per group). The pH values in the HC-treated group remained within the indicated range of 6.8–7.2, whereas those in the untreated control were generally below the lower limit of 6.8.

Figure 6 demonstrates that pH remained stable throughout the 14-day digestion period, with only minor daily fluctuations observed in both treatments. The coefficient of variation was lower in the HC-treated reactors (0.59%) than in the control (0.74%), indicating lower pH variability following HC pretreatment.

These findings suggest that HC enhanced methane production without causing process acidification or destabilization of the anaerobic digestion system. The absence of substantial pH fluctuations indicates that the enhanced methane-production response following cavitation did not result in substantial process acidification under the investigated conditions.

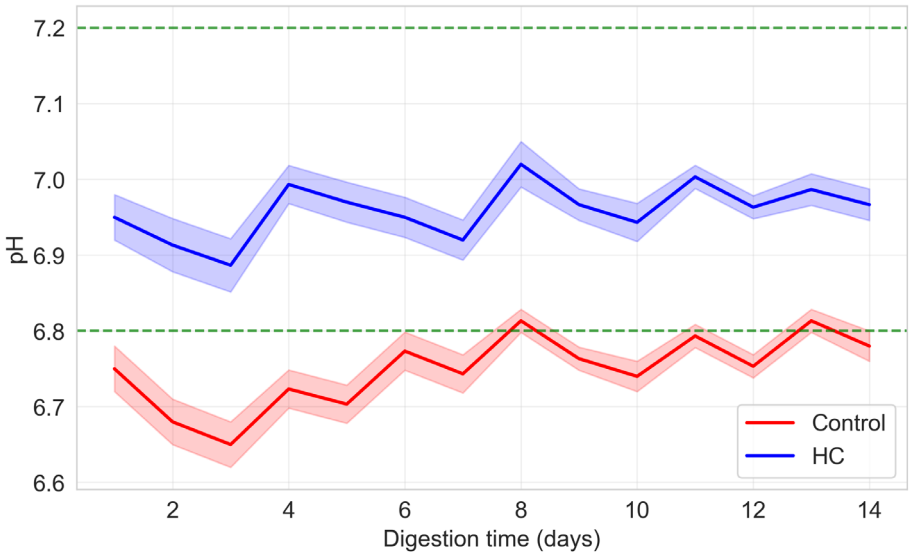


Figure 6. Daily pH profiles during batch anaerobic digestion of untreated and HC-pretreated cattle manure

Note: HC = hydrodynamic cavitation. Solid lines represent group means ($n = 3$), while shaded areas indicate the standard error of the mean (SEM). The dashed horizontal lines denote the recommended pH range (6.8–7.2) for stable methanogenic activity.

3.4 Biogas Production Performance

HC significantly enhanced cumulative biogas production throughout the anaerobic digestion process. At the end of the 14-day experiment, cumulative biogas production reached 497.7 ± 6.5 mL g^{-1} COD in the HC-treated reactors, compared with 397.6 ± 7.1 mL g^{-1} COD in the untreated control, corresponding to a 25.2% increase in cumulative biogas yield. The difference between the experimental groups was statistically significant ($p < 0.001$).

As illustrated in Figure 7, cumulative biogas production increased steadily in both treatments during the active digestion phase. However, the HC-treated reactors consistently maintained higher cumulative biogas production than the untreated control throughout the experiment. The divergence between the two curves became apparent during the first few days of digestion, increased progressively, and stabilized toward the end of the experimental period.

The higher cumulative biogas yield obtained after HC pretreatment indicates improved substrate conversion efficiency and enhanced microbial utilization of biodegradable organic matter. These findings are consistent with the cumulative methane production results (Figure 4) and further demonstrate enhanced biogas-production performance following HC pretreatment.

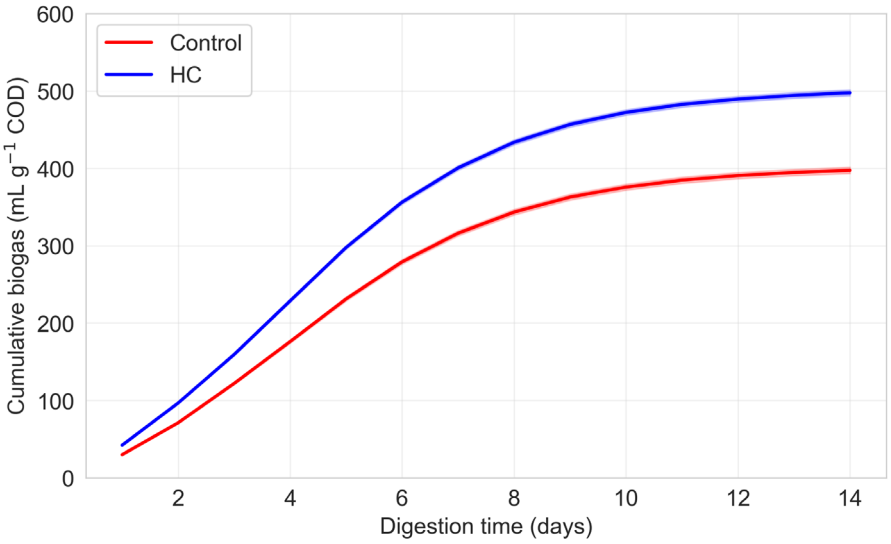


Figure 7. Cumulative biogas production during batch anaerobic digestion of untreated and HC-pretreated cattle manure

Note: HC = hydrodynamic cavitation; COD = chemical oxygen demand. Solid lines represent group means (n = 3), and shaded regions indicate the standard error of the mean (SEM).

3.5 Digestate Nutrient Enhancement

HC pretreatment was associated with higher concentrations of the major plant nutrients in the resulting digestate. Nitrogen concentration increased from 1.79 ± 0.03% in the control digestate to 2.25 ± 0.06% in the HC-treated digestate. Similar increases were observed for phosphorus (expressed as P_2O_5; 0.96 ± 0.02% vs. 1.18 ± 0.04%) and potassium (expressed as K_2O; 1.27 ± 0.03% vs. 1.59 ± 0.04%), all expressed on a dry-weight basis.

All differences between the experimental groups were statistically significant ($p < 0.01$). As shown in Figure 8, the HC-treated digestate consistently exhibited higher measured concentrations of N, P_2O_5 and K_2O than the untreated control.

The observed increases represent higher nutrient concentrations in the digestate on a dry-weight basis following HC pretreatment. Their possible relationship with enhanced organic matter degradation and concentration of the mineral fraction is discussed in Section 4.4.

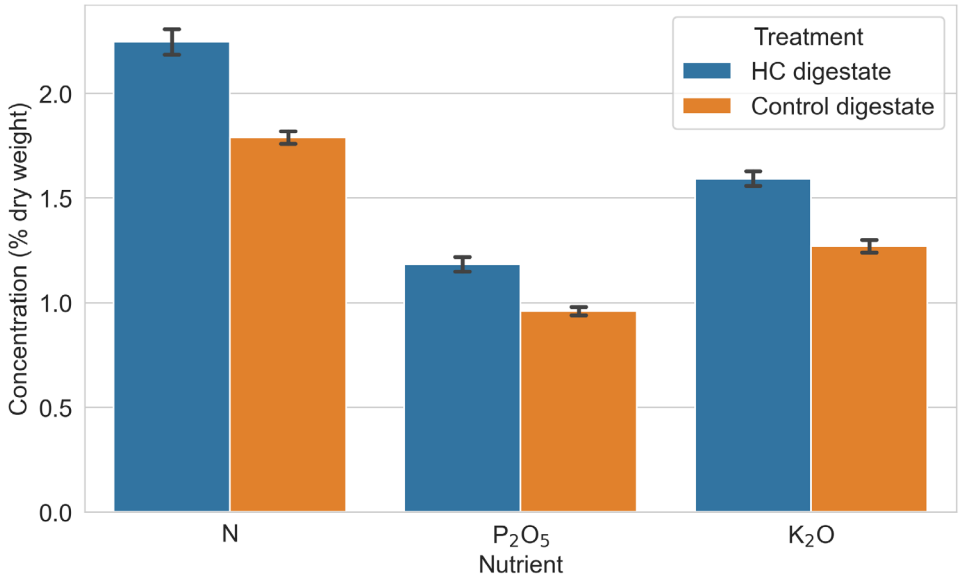


Figure 8. Concentrations of nitrogen (N), phosphorus (expressed as P_2O_5), and potassium (expressed as K_2O) in digestates obtained from untreated and HC-pretreated cattle manure

Note: HC = hydrodynamic cavitation. Nutrient concentrations are expressed on a dry-weight basis. Bars represent mean values (n = 3), and error bars indicate the standard deviation (SD).

3.6 Kinetic Modeling Using Modified Gompertz Equation

To quantitatively evaluate the effect of HC on methane production kinetics, cumulative methane production data were fitted using the modified Gompertz model. The model was used to estimate the methane production potential (P), maximum methane production rate (R_m), and lag phase (λ). The fitted curves are presented in Figure 9, and the estimated kinetic parameters together with the goodness-of-fit statistics are summarized in Table 4.

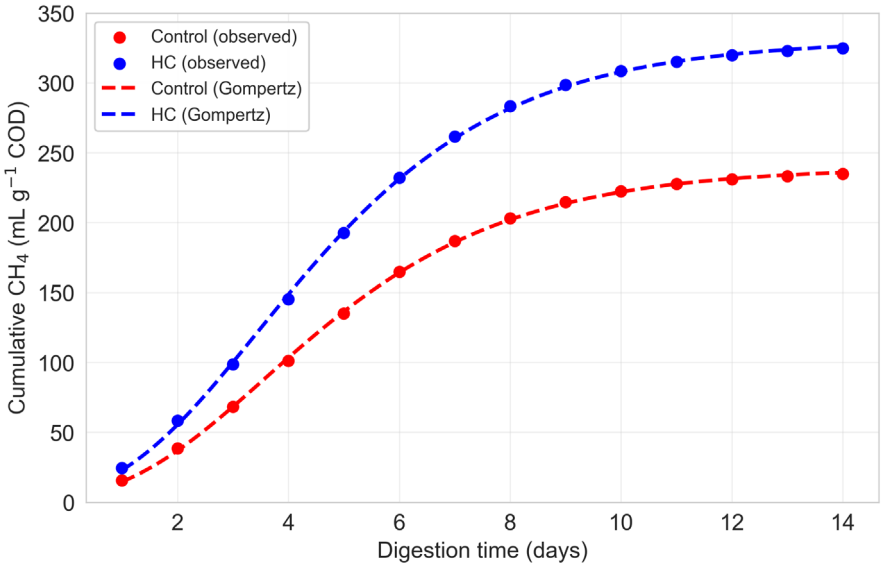


Figure 9. Modified Gompertz model fitting of cumulative methane production during batch anaerobic digestion of untreated and HC-pretreated cattle manure

Note: HC = hydrodynamic cavitation; COD = chemical oxygen demand. Symbols represent the experimental mean values (n = 3), whereas dashed lines indicate the fitted modified Gompertz model.

Table 4. Modified Gompertz kinetic parameters and model goodness-of-fit

Parameter	Control	HC-Treated
Methane production potential (P) (mL g^{-1} COD)	239.5	330.8
Maximum methane production rate (R_m) (mL g^{-1} COD d^{-1})	35.3	48.9
Lag phase (λ) (days)	1.07	0.96
R^2	0.9998	0.9997
RMSE (mL g^{-1} COD)	1.05	1.62

Note: HC = hydrodynamic cavitation; COD = chemical oxygen demand; RMSE = root mean square error.

The modified Gompertz model accurately reproduced the cumulative methane production profiles for both experimental groups, as evidenced by the excellent goodness-of-fit statistics ($R^2 > 0.999$ and RMSE < 2 mL g^{-1} COD). The close agreement between the experimental observations and the fitted curves confirms that the modified Gompertz equation adequately describes methane production kinetics under the investigated operating conditions.

HC substantially increased the estimated methane production potential from 239.5 to 330.8 mL g^{-1} COD, corresponding to an increase of approximately 38.1%. Likewise, the maximum methane production rate increased from 35.3 to 48.9 mL g^{-1} COD d^{-1}, indicating accelerated methane production following cavitation pretreatment.

The estimated lag phase was relatively short in both experimental groups, amounting to 1.07 days for the untreated control and 0.96 days for the HC-treated substrate. Although the reduction in lag phase was modest, the higher values of P and R_m indicate that the principal effect of HC was the enhancement of substrate biodegradability and methane production kinetics rather than a substantial shortening of the microbial adaptation period.

Overall, the kinetic analysis demonstrates that HC primarily increased methane production capacity and production rate while maintaining rapid microbial adaptation, thereby improving the overall efficiency of anaerobic digestion.

3.7 Energy Performance Assessment

The energy performance of HC pretreatment was evaluated by comparing the additional methane energy

recovered with the additional electrical energy required for cavitation. Electrical energy consumption was obtained from the PLC–SCADA energy inventory, while methane energy recovery was calculated from the experimentally determined increase in methane yield and the measured substrate COD.

The control configuration consumed 3.05 kWh of electrical energy per processing cycle, whereas the HC configuration required 4.96 kWh owing to the additional operation of the gerotor pump and cavitation disperser. Consequently, the additional electrical energy required for HC was 1.91 kWh run^{-1}.

Each pretreatment cycle processed 0.1005 m^3 of cattle manure slurry with an initial COD concentration of approximately 48 g L^{-1}. Based on the experimental methane yields obtained on day 14, HC increased cumulative methane production by 89.8 mL g^{-1} COD compared with the untreated control. Using this approximate COD concentration, this corresponded to an additional methane production of 4.31 m^3 CH_4 m^{-3} substrate.

Methane energy recovery was calculated using the LHV of methane (9.97 kWh m^{-3}, equivalent to 35.9 MJ m^{-3}). Accordingly, the additional methane produced by HC corresponded to an energy recovery of $\Delta E_{CH_4} = 4.31\ m^3\ CH_4\ m^{-3}_{substrate} \times 9.97\ kWh\ m^{-3}_{CH_4} = 42.97\ kWh\ m^{-3}_{substrate}$. The specific additional electrical energy input required for cavitation pretreatment was $\Delta E_{input} = \frac{1.91\ kWh}{0.1005\ m^3} = 19.00\ kWh\ m^{-3}_{substrate}$. The incremental NER was therefore calculated as $NER_{incremental} = \frac{42.97}{19.00} = 2.26$.

An incremental NER greater than unity indicates that the additional methane energy recovered exceeded the additional electrical energy required for HC. Under the investigated operating conditions, each unit of additional electrical energy supplied to the cavitation system generated approximately 2.26 units of recoverable methane energy.

These results demonstrate that HC enhanced methane production while maintaining a positive net energy balance. Although cavitation increased the electrical energy demand of the pretreatment stage, the additional methane energy recovered more than compensated for this energy input. Further improvements in pump efficiency, cavitation reactor geometry, and continuous-flow operation could further increase the overall energy efficiency and improve the techno-economic feasibility of HC pretreatment for farm-scale anaerobic digestion.

4. Discussion

4.1 Effect of Cavitation on Methane Production Kinetics

The obtained results demonstrate that HC substantially enhanced methane production during the anaerobic digestion of cattle manure. Compared with the untreated control, cavitation pretreatment increased cumulative methane production by approximately 38.2%, from 234.9 ± 5.9 to 324.7 ± 7.0 mL g^{-1} COD, while consistently increasing daily methane production throughout the active digestion period (Figure 5). These results indicate that cavitation pretreatment enhanced substrate conversion and methane production under the investigated conditions.

The observed enhancement is consistent with the current understanding that hydrolysis is a major rate-limiting stage of anaerobic digestion for particulate substrates such as cattle manure. Improving substrate accessibility may therefore contribute to increased methane productivity, as emphasized in recent reviews (Uddin & Wright, 2023). HC may promote substrate disintegration through the collapse of vapor bubbles, which generates localized pressure pulses, microjets, and shear forces capable of disrupting particulate organic matter and increasing the surface area accessible to hydrolytic microorganisms. However, because direct indicators of hydrolysis, such as soluble COD or particle-size distribution, were not measured in the present study, this mechanism should be regarded as a mechanistic interpretation rather than direct experimental confirmation.

The modified Gompertz model further supported the beneficial effect of cavitation pretreatment on methane production kinetics (Table 4 and Figure 9). HC increased the methane production potential from 239.5 to 330.8 mL g^{-1} COD and the maximum methane production rate from 35.3 to 48.9 mL g^{-1} COD d^{-1}. In contrast, the estimated lag phase changed only slightly, from 1.07 days for the control to 0.96 days for the HC-treated substrate, suggesting that the principal effect was associated with enhanced methane production kinetics rather than a substantial shortening of microbial adaptation. The agreement between the experimental data and the modified Gompertz model was high ($R^2 > 0.999$; RMSE < 2 mL g^{-1} COD).

The present findings are consistent with previous investigations showing that HC can enhance anaerobic digestion performance for livestock residues and other organic substrates (Dębowski et al., 2024; Langone et al., 2018; Patil et al., 2016). To place the obtained results into a broader engineering context, the increases in methane and biogas yields achieved in the present study were compared with those reported in selected original experimental studies (Figure 10). Because substrate composition, pretreatment technology, reactor configuration, treatment intensity, and experimental scale differed substantially among studies, this comparison is intended to provide contextual benchmarking rather than a direct ranking of pretreatment effectiveness.

The magnitude of methane enhancement observed in the present study was within the broad range reported for HC pretreatment, although substantial differences were evident among studies. Patil et al. (2016) reported maximum increases of approximately 145.0% in methane production and 88.5% in biogas production during rotor–

stator HC pretreatment of wheat straw. In contrast, Langone et al. (2018), using swirling-jet HC pretreatment of cattle manure followed by laboratory batch BMP assays, reported comparatively modest maximum increases of 3.1% in methane yield and 5.9% in biogas yield. These maxima were obtained under different operating pressures (7 bar for methane and 8 bar for biogas) and therefore should not be interpreted as simultaneous responses under a single operating condition.

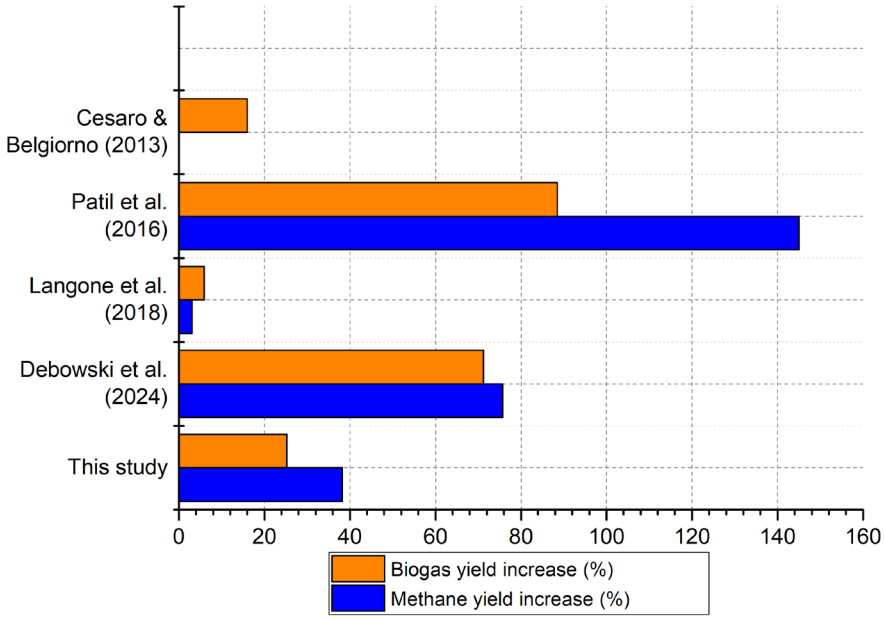


Figure 10. Comparison of reported increases in biogas and methane yields across selected pretreatment studies and the present study

Note: HC = hydrodynamic cavitation. Percentage increases are relative to the corresponding untreated controls. Differences in substrates, pretreatment methods, and operating conditions limit direct comparability. Missing bars indicate unavailable data.

At a larger scale, Dębowski et al. (2024) reported increases of 75.7% in methane production and 71.2% in biogas yield under the optimized full-scale V2 condition with an HC treatment duration of 8 min. These values refer specifically to the full-scale stage of that study and should not be confused with the larger responses observed during its laboratory-scale experiments.

For comparison with an alternative physical pretreatment technology, Cesaro & Belgiorno (2013) reported a 16% increase in biogas production following ultrasonic sonolysis of source-separated organic fraction of municipal solid waste. Because that study did not report a corresponding increase in methane yield, no methane value is included in Figure 10. Sonolysis is included only as a contextual comparison and should not be interpreted as HC. Under the pilot-scale conditions investigated in the present study, Venturi-based HC increased cumulative methane yield by 38.2% and cumulative biogas production by 25.2%.

The differences among studies can be attributed, at least in part, to variations in substrate characteristics, pretreatment configuration, operating conditions, and experimental scale. Wheat straw is a recalcitrant lignocellulosic substrate in which physical disruption may substantially affect substrate accessibility, whereas cattle manure has already undergone partial biological degradation in the animal digestive tract. The full-scale system investigated by Dębowski et al. (2024) also processed a mixture of dairy cattle slurry and milking wastewater under operating conditions substantially different from those used in the present pilot-scale system.

Energy performance likewise differed among the studies. Langone et al. (2018) reported net energy yields approximately 17–25% lower than that of the untreated control under the investigated HC conditions, whereas Dębowski et al. (2024) reported an approximately 3.1% increase in net energy output under the selected full-scale condition. In the present study, the incremental NER was 2.26, indicating that the additional recoverable methane energy exceeded the additional electrical energy required specifically for HC pretreatment. These energy indicators should not be directly compared or ranked because the studies used different system boundaries, operating scales, substrates, and energy-accounting methodologies.

Overall, the benchmarking analysis indicates that the pilot-scale HC pretreatment investigated in the present study produced substantial increases in methane and biogas production under the tested conditions. Together with the positive incremental energy ratio, these results support further evaluation of Venturi-based HC for cattle-manure treatment.

4.2 Cavitation Operating Conditions and Mechanistic Interpretation

From an engineering perspective, the higher methane production rate together with the increased methane production potential may improve reactor productivity and enhance process efficiency without compromising microbial stability. For organic farming applications, more efficient stabilization of cattle manure can contribute to improved renewable energy recovery while simultaneously producing nutrient-rich digestate suitable for circular nutrient management.

The enhancement of methane production following HC pretreatment may be partly attributed to intensification of the hydrolysis stage, which is generally recognized as the rate-limiting step during anaerobic digestion of particulate substrates such as cattle manure. The higher methane production potential and maximum methane production rate obtained in the present study are consistent with improved substrate biodegradability, potentially associated with physical disintegration of particulate organic matter (Table 4 and Figure 9).

The intensity of HC is fundamentally governed by the cavitation number, defined as

$$\sigma = \frac{P_{\text{out}} - P_{\text{vapor}}}{\frac{1}{2}\rho v_{\text{throat}}^2} \tag{4}$$

where, P_{out} is the downstream (outlet) pressure, P_{vapor} is the vapor pressure of the liquid, ρ is the fluid density, and v_{throat} is the fluid velocity at the Venturi throat.

In the present study, the downstream (outlet) pressure was selected as the reference pressure because it directly characterizes the pressure recovery region where cavitation bubbles collapse after passing through the Venturi throat. Under the investigated hydraulic configuration, bubble collapse rather than bubble inception governs the intensity of cavitation-induced mechanical effects responsible for biomass disintegration. Therefore, the outlet pressure provides a practical representation of cavitation intensity within the pilot-scale reactor and was considered the most appropriate reference parameter for calculating the cavitation number.

The selected operating conditions (inlet pressure 4.3–4.8 bar, outlet pressure approximately 0.43 bar, and pressure drop 3.9–4.1 bar) produced a pressure ratio of approximately 10:1, which is characteristic of Venturi reactors capable of generating intense cavitation while maintaining stable hydraulic performance. Under these conditions, the collapse of cavitation bubbles may generate localized pressure pulses, microjets, and high shear stresses capable of fragmenting manure particles and partially disrupting cell structures, thereby potentially increasing the surface area available for hydrolytic microorganisms and enzymes.

The experimental observations are consistent with this mechanistic interpretation. HC increased the methane production potential from 239.5 to 330.8 mL g^{-1} COD and the maximum methane production rate from 35.3 to 48.9 mL g^{-1} COD, whereas the lag phase remained relatively short and changed only slightly (1.07 days for the untreated control and 0.96 days for the HC-treated substrate). These results are consistent with the interpretation that cavitation primarily enhanced substrate accessibility and biodegradability rather than substantially accelerating microbial adaptation. Likewise, the consistently higher daily methane production observed during the active digestion stage (Figure 5) is consistent with improved substrate conversion, potentially associated with enhanced hydrolysis and increased availability of soluble intermediates for methanogenesis.

The increase in digestate nutrient concentrations observed after cavitation pretreatment is also consistent with improved disintegration and solubilization of particulate organic matter. Higher concentrations of nitrogen, phosphorus and potassium in the digestate suggest that cavitation enhanced the release of nutrients associated with particulate biomass, although the specific mechanisms of nutrient release were not directly investigated in the present study. Consequently, this interpretation should be regarded as a plausible mechanistic explanation rather than direct experimental evidence.

The energy analysis further indicates that enhanced methane production was achieved with a positive incremental energy balance under the investigated conditions. The additional electrical energy required for HC corresponded to 19.0 kWh m^{-3} of processed substrate, whereas the additional methane produced provided 42.97 kWh m^{-3} of recoverable energy, resulting in an incremental NER of 2.26. These results indicate that the additional energy required for cavitation pretreatment was more than compensated by the increased methane production, supporting the practical applicability of Venturi-based HC under pilot-scale operating conditions.

The present observations are consistent with previous studies reporting that HC enhances hydrolysis by disrupting particulate organic matter, increasing substrate accessibility and facilitating subsequent microbial degradation (Dular et al., 2016; Ge et al., 2021; Gogate & Pandit, 2001; Langone et al., 2018). Since hydrolysis is generally considered the principal rate-limiting step during anaerobic digestion of lignocellulosic and particulate substrates, improved substrate solubilization can substantially accelerate downstream acidogenic and methanogenic processes. Improved hydrolysis may also influence acetate conversion pathways and syntrophic microbial interactions, which are increasingly recognized as important determinants of methane production efficiency in anaerobic digestion systems (Pan et al., 2021).

4.3 Process Stability and Methanogenic Conditions

Despite the substantial enhancement of methane production, HC pretreatment did not cause substantial pH destabilization during anaerobic digestion. Throughout the 14-day experimental period, the HC-treated reactors maintained pH values within the indicated favorable range of 6.8–7.2, whereas the untreated control generally exhibited slightly lower pH values.

Stable pH is essential for maintaining balanced microbial activity during anaerobic digestion because methanogenic archaea are particularly sensitive to excessive acidification. In the present study, the average pH after feeding was 6.75 ± 0.02 in the untreated control and 6.96 ± 0.02 in the HC-treated reactors. Although the numerical difference was relatively small, it was statistically significant and was accompanied by lower pH variability in the cavitation-treated group (coefficient of variation 0.59% compared with 0.74% for the control).

As illustrated in Figure 6, the HC-treated reactors maintained consistently narrower pH fluctuations throughout the digestion period than the untreated control. These observations suggest that HC did not destabilize pH during digestion despite the enhanced methane production. Instead, cavitation pretreatment maintained favorable pH conditions while improving methane production performance.

The slightly higher average pH observed after cavitation pretreatment may be related to physicochemical changes occurring during HC. Previous studies have suggested that intense pressure fluctuations and bubble collapse may promote partial removal of dissolved carbon dioxide from the liquid phase, thereby slightly increasing pH (Langone et al., 2018; Ranade, 2022). Although dissolved CO_2 and alkalinity were not measured in the present study, the observed pH increase is consistent with this proposed mechanism and therefore represents a plausible explanation rather than direct experimental evidence.

The maintenance of stable pH throughout the experiment suggests that no substantial pH destabilization occurred under the investigated conditions. While volatile fatty acids were not directly quantified in this study, the absence of a marked downward pH trend suggests that no substantial pH destabilization occurred under the selected operating conditions.

Overall, these results demonstrate that HC intensified substrate biodegradation and methane production without compromising process stability. The combination of enhanced methane production, stable pH conditions and low pH variability indicates that the selected cavitation operating regime was compatible with sustained methanogenic activity and reliable anaerobic digestion performance.

4.4 Digestate Quality and Nutrient Availability

In addition to enhancing methane production, HC influenced the composition of the resulting digestate. Compared with the untreated control, cavitation pretreatment increased the measured concentrations of nitrogen, phosphorus and potassium on a dry-weight basis. Nitrogen concentration increased from 1.79 ± 0.03% to 2.25 ± 0.06%, phosphorus (expressed as P_2O_5) increased from 0.96 ± 0.02% to 1.18 ± 0.04%, and potassium (expressed as K_2O) increased from 1.27 ± 0.03% to 1.59 ± 0.04%. These changes indicate a higher measured nutrient content in the digestate after pretreatment.

The observed increase in nutrient concentrations should be interpreted as an increase in the measured nutrient content rather than direct evidence of increased nutrient bioavailability. Because nutrient fractionation was not performed, the proportions of plant-available nitrogen, phosphorus and potassium were not determined. Future studies should therefore include NH_4^+–N, soluble phosphorus and exchangeable potassium, together with electrical conductivity, salinity and agronomic testing.

HC may contribute to these compositional changes through physical disruption of manure particles and partial disintegration of microbial and plant cell structures. Such mechanical effects increase the contact area between particulate organic matter and anaerobic microorganisms, facilitating the release of nutrients associated with particulate biomass into the digestate matrix. However, the specific mechanisms governing nutrient transformation and partitioning were not directly investigated in the present study and therefore require further research.

The higher nutrient concentrations observed in the HC-treated digestate may partly reflect compositional changes associated with conversion of organic matter during anaerobic digestion. However, because a complete nutrient mass balance and nutrient fractionation were not performed, the underlying mechanisms cannot be quantified from the present data. As biodegradable solids were converted into biogas, the remaining digestate became relatively enriched in mineral nutrients, which is consistent with the higher concentrations of nitrogen, phosphorus and potassium measured after cavitation pretreatment.

It should be emphasized that the total mass of nitrogen, phosphorus and potassium is expected to remain approximately conserved during anaerobic digestion. Therefore, the higher nutrient concentrations observed in the HC-treated digestate may partly reflect the conversion of organic matter into biogas and the resulting concentration of nutrients within the remaining solid fraction, rather than the formation of additional nutrients. Accordingly, the observed increases should be interpreted as concentration effects on a dry-weight basis and not as direct evidence of increased nutrient production or bioavailability.

From an agricultural perspective, improved digestate composition increases the potential value of anaerobic digestion as both a renewable energy technology and a nutrient recovery process. In organic farming systems, these benefits are particularly important because livestock manure is regarded as a valuable resource rather than a waste product. The simultaneous recovery of renewable energy and the potential recycling of nutrients through digestate application contributes to circular nutrient management, although additional agronomic studies are required to verify nutrient availability and fertilizer performance under field conditions.

These findings are consistent with the principles of the circular bioeconomy, in which livestock manure is simultaneously utilized for renewable energy production and recovery of nutrient-containing digestates suitable for further agronomic evaluation. Future studies should therefore complement chemical characterization of digestates with nutrient fractionation, determination of NH_4^+–N, soluble phosphorus and exchangeable potassium, as well as assessments of electrical conductivity, salinity, heavy metals, pathogen reduction and agronomic performance under field conditions (Möller & Müller, 2012; Nkoa, 2014).

4.5 Energy Performance and Industrial Implications

The energy assessment demonstrated that HC increased the electrical energy demand of the pretreatment stage; however, the additional methane produced during anaerobic digestion more than compensated for this increase in energy consumption. Based on the experimentally determined methane yield, substrate COD, and the LHV of methane (9.97 kWh m^{-3} CH_4, equivalent to 35.9 MJ m^{-3}), HC generated an additional 4.31 m^3 CH_4 m^{-3} of processed substrate. This corresponded to an additional recoverable methane energy of 42.97 kWh m^{-3}.

The electrical energy inventory obtained from the PLC–SCADA monitoring system showed that the control configuration consumed 3.05 kWh per processing cycle, whereas the HC configuration required 4.96 kWh owing to the additional operation of the gerotor pump and cavitation reactor. Consequently, the additional electrical energy required for cavitation pretreatment was 1.91 kWh per processing cycle, corresponding to a specific energy input of 19.0 kWh m^{-3} of processed substrate.

Comparison of the additional methane energy recovered with the additional electrical energy required exclusively for HC pretreatment yielded an incremental NER of 2.26. It should be emphasized that this incremental NER refers exclusively to the HC pretreatment stage and should not be interpreted as the overall energy balance of the complete anaerobic digestion process. Energy requirements associated with substrate heating, digester operation, mixing, biogas upgrading, gas compression, and downstream utilization were intentionally excluded from the present system boundary. Consequently, the reported incremental NER should be interpreted exclusively as a performance indicator of the HC pretreatment stage under the investigated pilot-scale conditions and not as the overall net energy balance of a complete anaerobic digestion facility.

This value indicates that each unit of additional electrical energy consumed by the cavitation system generated approximately 2.26 units of recoverable methane energy. Although direct comparison with previously reported estimates is limited by differences in system boundaries and calculation methods, the results indicate a positive incremental energy balance for the HC pretreatment stage under the investigated pilot-scale conditions.

The positive incremental energy balance obtained in the present study indicates that HC may be energetically favorable under the investigated conditions. Differences between the present results and previously published studies are expected because energy performance depends strongly on substrate characteristics, reactor geometry, cavitation intensity, operating pressure, hydraulic configuration and the methodology used for energy accounting. In the present work, electrical energy consumption was determined from the pilot-scale energy inventory, whereas methane energy recovery was estimated from laboratory-scale methane yields and substrate COD measurements, providing a transparent and reproducible basis for evaluating incremental energy performance.

Although the cavitation system required additional electrical energy, further improvements in energy efficiency remain feasible. Optimization of Venturi geometry, reduction of unnecessary pressure losses, improvement of pump hydraulic efficiency and optimization of treatment duration may further decrease the specific electrical energy demand without compromising cavitation intensity. Moreover, continuous-flow operation and industrial-scale implementation are expected to improve hydraulic efficiency and reduce specific energy consumption through economies of scale.

From a practical implementation perspective, the proposed HC system is particularly suitable for medium- and large-scale livestock farms where manure is already collected, homogenized, and mechanically transported prior to anaerobic digestion. The additional equipment required for cavitation pretreatment consists primarily of a Venturi reactor, a circulation pump, and a PLC–SCADA control unit, which can be integrated into existing manure handling systems without fundamentally changing the anaerobic digestion process. Nevertheless, successful industrial implementation will require optimization of electrical energy consumption, maintenance requirements, hydraulic reliability, and operating costs. Therefore, comprehensive techno-economic evaluation and long-term continuous pilot-scale operation remain essential before full-scale commercial deployment.

From an engineering perspective, the overall performance of HC should not be evaluated solely on the basis of electrical energy balance. The technology simultaneously enhanced methane production, increased methane

production kinetics, improved substrate biodegradability and produced a digestate with higher nutrient concentrations. These combined benefits may improve the overall technical and economic performance of anaerobic digestion systems, particularly in integrated organic farming, where renewable energy generation and nutrient recycling are complementary objectives.

Consequently, evaluation of HC should consider both energy and non-energy benefits within a broader circular bioeconomy framework. Future studies should combine techno-economic analysis, life-cycle assessment and long-term pilot-scale validation to determine the overall sustainability and economic feasibility of cavitation-assisted anaerobic digestion under commercial farm conditions.

4.6 Study Limitations and Future Research Directions

Although the present study indicates that pilot-scale HC pretreatment can enhance methane production in subsequent laboratory-scale batch digestion, several limitations should be considered when interpreting the results.

First, HC pretreatment was conducted using a pilot-scale system, whereas anaerobic digestion was evaluated through laboratory-scale batch experiments. Although the pilot-scale pretreatment system provides practical engineering information, the digestion results cannot be directly extrapolated to full-scale continuous anaerobic digesters.

Second, several important physicochemical parameters, including volatile fatty acids, alkalinity, dissolved carbon dioxide, rheological properties and soluble nutrient fractions, were not measured. Consequently, the proposed mechanisms of enhanced hydrolysis, partial CO_2 stripping, buffering behavior and nutrient solubilization should be regarded as mechanistic interpretations supported by indirect evidence rather than direct experimental confirmation.

Third, although the modified Gompertz model accurately described cumulative methane production, the model represents an empirical kinetic approximation and does not explicitly account for the individual biochemical stages of anaerobic digestion or the complex hydrodynamic phenomena occurring within the cavitation reactor. Likewise, the cavitation analysis was based primarily on the calculated cavitation number and measured operating parameters, without detailed computational fluid dynamics simulations of multiphase flow, bubble dynamics, or the non-Newtonian rheology of cattle manure.

Another limitation concerns the extent of inoculum characterization rather than the absence of inoculum information. In the present study, the origin of the inoculum, TS, VS, ISR, and the use of blank assays were fully documented and are described in Section 2.3. These experimental conditions ensured a consistent comparison between the untreated control and HC-treated reactors. However, additional inoculum characteristics, including specific methanogenic activity and microbial community composition, were not determined. Consequently, although the experimental design satisfies the objectives of the present comparative study, future investigations would benefit from more comprehensive inoculum characterization to further improve reproducibility, facilitate comparison among different anaerobic digestion systems, and strengthen the interpretation of microbial process dynamics.

The present energy assessment was based on experimentally measured electrical energy consumption and methane production under the investigated operating conditions. Although the results demonstrated a positive incremental NER, additional optimization of Venturi geometry, operating pressure, treatment duration and pump hydraulic efficiency may further improve the overall energy performance of the cavitation system. Future studies should therefore investigate continuous-flow operation, long-term pilot-scale performance and industrial-scale implementation.

Future research should also include comprehensive characterization of digestate quality, including nutrient fractionation, NH_4^+–N, soluble phosphorus, exchangeable potassium, pathogen reduction, heavy metals and agronomic performance under field conditions. Integration of computational fluid dynamics, microbial community analysis, techno-economic assessment and life-cycle assessment would further strengthen the understanding of HC as a pretreatment technology for sustainable anaerobic digestion within circular bioeconomy and organic farming systems.

5. Conclusions

The present study demonstrated that pilot-scale HC pretreatment effectively enhanced methane production during laboratory-scale batch anaerobic digestion of cattle manure. Compared with the untreated control, cavitation pretreatment increased cumulative methane production from 234.9 ± 5.9 to 324.7 ± 7.0 mL g^{-1} COD, corresponding to an increase of approximately 38.2%. Daily methane production was consistently higher throughout the active digestion period, consistent with enhanced methane-production kinetics and substrate conversion under the investigated conditions. Kinetic analysis using the modified Gompertz model confirmed that HC increased both the methane production potential (239.5 → 330.8 mL g^{-1} COD) and the maximum methane production rate (35.3 → 48.9 mL g^{-1} COD d^{-1}), whereas the lag phase remained relatively short in both treatments

(1.07 and 0.96 days). These findings are consistent with HC improving substrate accessibility and potentially intensifying hydrolysis, while the lag phase changed only slightly. Despite the higher methane production rate, cavitation pretreatment maintained relatively stable pH conditions throughout the digestion process. The HC-treated reactors maintained average pH values within the favorable range for methanogenesis (6.96 ± 0.02), whereas the untreated control exhibited a slightly lower mean pH (6.75 ± 0.02). The lower pH variability observed after cavitation further indicated more stable pH conditions under the selected operating conditions. HC also improved digestate composition by increasing the measured concentrations of nitrogen, phosphorus and potassium on a dry-weight basis. Although nutrient bioavailability was not directly evaluated, the improved nutrient profile suggests enhanced potential for nutrient recycling through digestate utilization. These findings are particularly relevant for organic farming systems, where renewable energy generation and recycling of farm-derived nutrients represent complementary components of sustainable agricultural management. Overall, the results indicate that HC improved methane production and maintained a positive incremental energy balance under the investigated pilot-scale conditions. These findings support the potential of Venturi-based cavitation as a practical pretreatment strategy for enhancing anaerobic digestion performance in cattle manure systems. The integration of Venturi-based HC with PLC–SCADA process monitoring provides a potential approach for improving anaerobic digestion performance while maintaining stable pH conditions and a positive incremental energy balance under the investigated conditions. The combination of enhanced methane production, improved substrate biodegradability, favorable digestate composition and positive energy balance demonstrates the potential of HC as an effective pretreatment technology for livestock manure management. Future research should focus on optimization of Venturi reactor geometry, reduction of specific electrical energy consumption, continuous-flow operation, computational fluid dynamics modeling, long-term pilot-scale validation, and comprehensive techno-economic and life-cycle assessments. Additional studies should also investigate nutrient fractionation, microbial community dynamics and agronomic performance of the resulting digestate to further support industrial implementation of HC within circular bioeconomy and organic farming systems.

Author Contributions

Conceptualization, S.M.I. and M.A.B.; methodology, S.M.I., M.A.B., and M.B.T.; software, S.M.I. and R.N.T.; validation, B.I.B., M.B.T., and K.M.A.; formal analysis, S.M.I., R.N.T., and B.I.B.; investigation, S.M.I., M.A.B., and M.B.T.; resources, M.B.T. and K.M.A.; data curation, S.M.I. and R.N.T.; writing—original draft preparation, S.M.I. and M.A.B.; writing—review and editing, M.A.B., B.I.B., M.B.T., and K.M.A.; visualization, S.M.I. and R.N.T.; supervision, M.A.B., B.I.B., and M.B.T.; project administration, M.A.B. and K.M.A. All authors have read and agreed to the submitted version of the manuscript.

Data Availability

The data supporting the findings of this study are available from the corresponding author upon reasonable request.

Conflicts of Interest

The authors declare no conflicts of interest.

Declaration on the Use of Generative AI and AI-assisted Technologies

In preparing and revising this manuscript, the authors used OpenAI ChatGPT solely to assist with English-language editing and to improve the clarity and readability of the text. The AI tool was not used to generate experimental data or scientific results, perform the reported data analyses, fabricate references, or replace the authors' scientific judgment and interpretation. All scientific content, numerical results, references, interpretations, and conclusions were reviewed and verified by the authors, who take full responsibility for the content and final version of the manuscript.

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